Virtee Parekh

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WORK EXPERIENCE

Senior Machine Learning Engineer at Indeed.com Apr 2021 – Present

- Developed a learning to rank model that improved jobs quality on feed by 5% and reduced negative user engagement by 15%.
- Developing embedding features from a compact LLM (e.g. Qwen) and integrating them into a production ranking model to improve feature representation and model performance.
- Unified multiple production models into a single comprehensive model by applying targeted modeling optimizations to reduce training time and improve performance; cut tech debt, delivered a +2% model lift, and maintained user engagement.
- Implemented production level code to engineer multiple model features pivotal to ranking models' accuracy and performance.
- Led multiple experiments to iteratively enhance model accuracy by introducing new features, resulting in significant improvements in model performance.
- Collaborated with engineering teams to deliver science support in developing an innovative feature engineering process, reducing the feature engineering timeline from ~3 months to ~1 month.
- Designed and conducted experiments by implementing model architecture modifications, resulting in a reduction in training time by 80% without harming model accuracy.
- Improved model performance by 3% by developing and integrating complex multi-target models from single-target models.
- Implemented the code to incorporate a new ads bidding optimization technique for the jobs search ranking algorithm.
- Designed, executed and analyzed A/B tests to evaluate the performance of a new ranking algorithm, resulting in its rollout in 3 international markets.
- Conducted offline experiments to evaluate optimal features and hyper parameters for a learning to rank model, enhancing the model's predictive accuracy by 1.5%.
- Designed and conducted A/B tests to improve jobs search ranking by penalizing jobs of poorer quality.
- Performed power analysis to determine success KPIs for conducting experiments.
- Developed data artifacts that provide critical data inputs for a product's pricing algorithm.
- **Technical Stack:** Python, Java, Experimentation, Hypothesis Testing, Search Ranking, Machine Learning

Business Intelligence Analyst II at Indeed.com Feb 2019 – Apr 2021

- Built a scalable revenue forecasting pipeline using time series analysis that helps the senior leadership plan investments better.
- Performed regression analysis to determine leading indicators that are reliable signals of imminent changes to revenue.
- Reduced the error of a forecasting model across markets from 20% to 7% that has a revenue impact in hundreds of thousands.
- Developed a quota pacing dashboard for the Sales team using predictive modeling that saved resources worth millions.
- Built a machine learning model to predict the number of requisitions made that facilitate better decision making for Sales.
- Conducted causal impact analysis to determine the impact of accepting alternate payment methods on revenue.
- Performed survival analysis to develop an interpretable model that predicts lifetime value, affecting significant portions of the marketing budget.
- Led a machine learning study group for a team of 15+ members where I presented tutorials and curated learning resources.
- Winner of Product, Technology, and Engineering award in an Indeed-wide hackathon for developing a resume information extraction tool.
- **Technical Stack:** Python, R, SQL, Applied Data Science, Time Series Forecasting, Statistical Analysis, Tableau, AWS

Grader at Rochester Institute of Technology Jan 2018 – Dec 2018

- CSCI 720 - Big Data Analytics
- CSCI 420 - Principles of Data Mining

Business Intelligence Intern at Indeed.com May 2018 – Aug 2018

- Built a revenue forecasting model with an error rate of 5% that reduced manual effort by 1 hour daily.
- Performed exploratory data analysis to address an unsolved business problem of building a logic from complex source data.
- **Technical Stack:** Python, SQL, Facebook Prophet, Time Series Analysis and Forecasting

SKILLS

Programming/Scripting Languages	Python, R, Java
Databases/Query Engines	PostgreSQL, Snowflake, Presto, Hive
Data Science Tools	Scikit-learn, Tensorflow, PyTorch
Visualization Tools	Python, Tableau
Cloud Computing Platforms	Amazon Web Services

EDUCATION

Master of Science in **Computer Science**	Rochester Institute of Technology	Dec 2018
Bachelor of Engineering in **Computer Engineering**	University of Pune	May 2016

PROJECTS

Exploring Latent Cognitive Biases in Employer Compensation Aug 2018 – Dec 2018

- Performed exploratory data analysis to look for observable trends on companies that have a gender pay gap.
- Analyzed tweets related to the gender pay gap by performing sentiment analysis and generating word clouds.